

Nico Seamans

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EDUCATION

B.Sc. in Mechanical Engineering (25-26 Dean's List GPA), Minor in Mathematics

University of New Brunswick

Dec 2027 graduation

Fredericton, New Brunswick

WORK EXPERIENCE

Aerospace Manufacturing Engineering Intern

Pratt & Whitney

Jan 2026 – Aug 2026

Enfield, Nova Scotia

- Designed and built an airfoil lab fixture for CMM inspection to validate machining tolerance in accordance with GD&T specifications, reducing setup-induced measurement error and improving inspection speed and accuracy by ~10%
- Designed 3D-printed polishing templates in CATIA V5 to support Design for Manufacturing (DFM), accelerating manual polishing on IBF fan blades by ~25% (trailing edge) and 14% (leading edge)
- Developed manufacturing processes and documentation, including process flows, machine utilization, and basic NC programming, for the Military Collaborative Combat Aircraft (CCA) Project; led meetings and presented updates to engineers and senior managers
- Partnered with engineers, fellows, and senior managers to specify and source a 5-axis mill-turn machine for projected production volume increases and new program requirements

Co-Founder & Senior Project Lead

UNB Formula SAE EV

Oct 2025 – Present

Fredericton, New Brunswick

- Co-leading the dynamics team of 30+ students in the Formula SAE EV program, coordinating cross-functional timelines toward the team's first competition
- Leading Manufacturing for the team, directing fixture and jig design and establishing process standards and work instructions for custom part machining to enable high-precision, repeatable manufacturing of critical components
- Designed a modular jig table with a standardized mounting interface, enabling 3D-printed jig inserts to be swapped in for custom fixturing at a fraction of the traditional tooling cost.
- Built early quarter- and half-car models in MATLAB/Simscape to guide suspension geometry and validate handling targets

Mechanical Engineering Student

J.D. Irving Limited

May 2025 – Dec 2025

Chipman, New Brunswick

- Led design and installation of a remote-controlled gantry crane spanning seven log beds, applying root-cause analysis to cut operator exposure and reduce maintenance downtime
- Designed a two-stage winch recovery system to extract trapped logs from woodchipper operations, implementing a corrective action that removed the need for manual entry
- Managed design and installation of a retrofit fall-protection system, including a crossbeam tie-off structure, improving safety compliance for millwright operations

TECHNICAL SKILLS

CAD & Engineering: CATIA V5, Siemens NX, SolidWorks, Fusion 360, Technical Drawings, FEA (ANSYS)

Manufacturing: DFM, GD&T, Fixture Design, FAI, Lean Six Sigma White Belt, Manufacturing Processes, SAP, Enovia

Analysis & Programming: MATLAB, Simulink, Simscape, Excel, Python

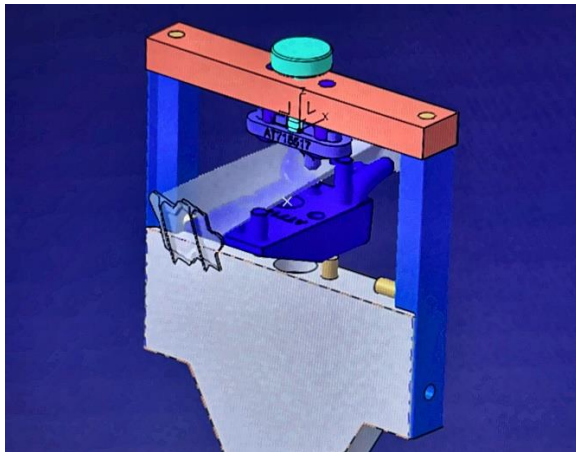
EXTRACURRICULARS

- Team Canada Powerlifter: 2025 World Bench Press Bronze Medallist (Drammen, Norway); 2024 Best U24 Bench Presser in Canada; 2025 New Brunswick Powerlifting Best U24 Powerlifter
- NBPL Chair & Board Member
- 2024 Atlantic Engineering Competition: Junior Design (presented design to judges)

Portfolio -Nico Seamans

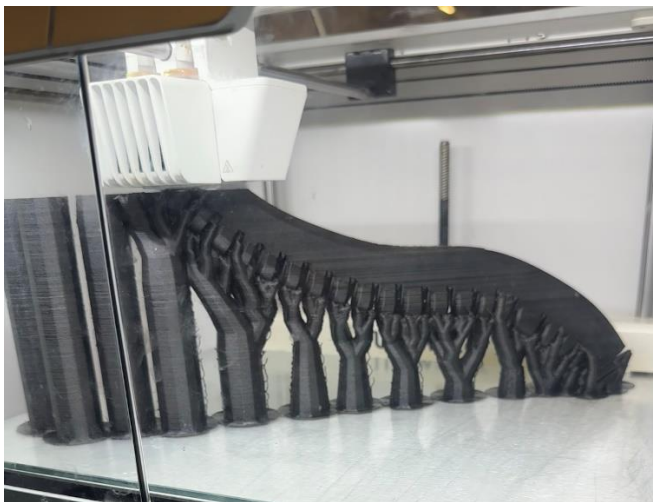
Coordinate Measure Machine (CMM) Airfoil Inspection Fixture

- Designed a fixture to improve CMM measurement consistency when inspecting blades that did not meet tolerance during manufacturing operations.
- Revised the existing blade-holding fixture to move the balance point toward the blade root, letting the CMM inspect a larger blade surface area without colliding with the fixture.
- Calculated the force each side magnet needed to apply optimal pressure on the blade for consistent fixture placement.
- Created a DFM design so each component could be manufactured on a wire EDM for tight tolerances.



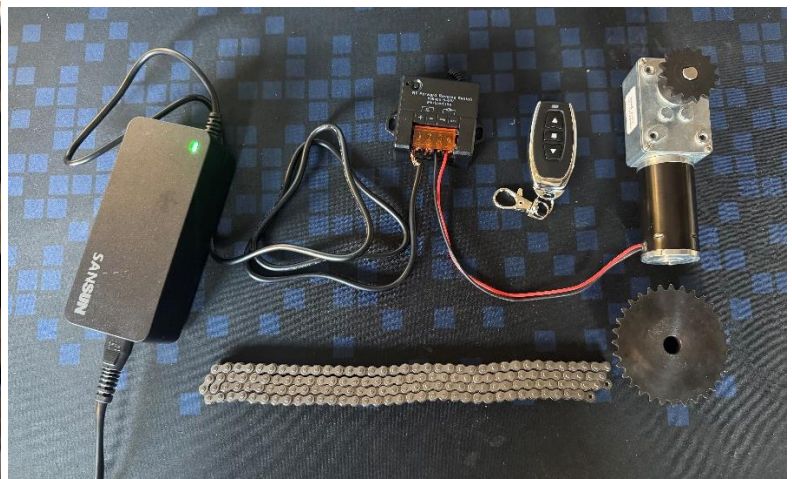
Integrally Bladed Fan (IBF) Polishing Template

- Created a manual polishing gauge to troubleshoot a batch with miscorrelation between auto-polishing, CMM results, and visual inspection.
- 3D printed rapid prototypes of the leading- and trailing-edge templates that follow the blade profile and include marking lines for visual correlation with CMM results.
- Troubleshot inspection data to justify a CMM process change, slowing the probe speed to avoid potential deflection.
- Designed an end-of-blade locating feature so the template lines up consistently, eliminating operator error.



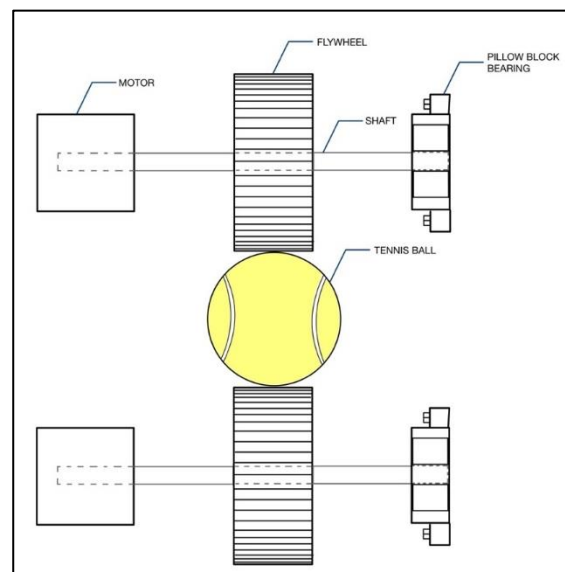
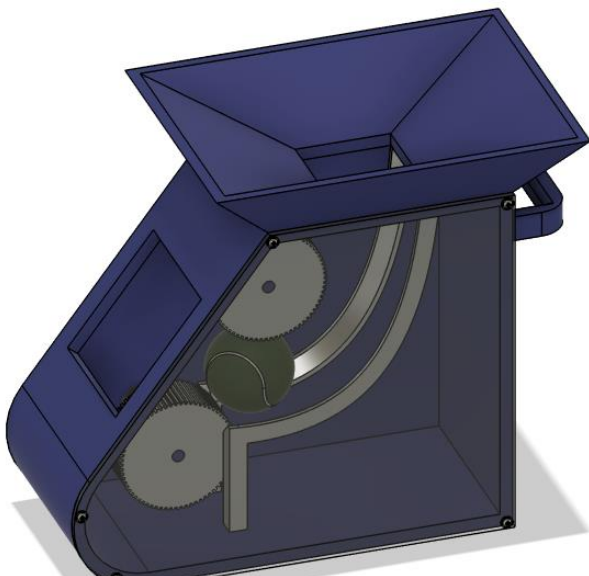
Indoor Golf Simulator and Retractable Impact Screen

- Designed and built a remote-controlled retractable impact screen for a golf simulator for under \$500.
- Built the frame from 2 in ID conduit, a garage door opener rack, ball bearings, 10 mm bolts, fishing net, and a triple-layer impact screen.
- Drove it with a reversible 12 V worm-gear motor, chosen for high torque in a compact package.
- Geared it with a 15-tooth sprocket on the 8 mm motor shaft driving a 32-tooth sprocket through #25 roller chain, for about 2.1:1 additional reduction.
- Powered it from a 12 V, 5 A supply, with a 50 m wireless relay module that switches motor direction, so the screen raises and lowers from a remote.



Dual-Flywheel Dog Ball Launcher (ME 3345 Machine Design Project)

- Designed a dual-flywheel tennis ball launcher in CAD, with a gravity-fed hopper and an enclosed wheel housing, for consistent launch trajectories.
- Verified the flywheel, shaft, key, and bearing designs by hand calculation, with fatigue safety factors from 3.2 to 30 and a bearing life of 2.4×10^{11} revolutions.
- Specified a fused, switched 3S (11.1 V) LiPo battery feeding two brushless motors, each driven by its own ESC (electronic speed controller).
- Controlled both ESCs from a single servo tester signal through a splitter, so one knob sets both wheel speeds.



Department of Mechanical Engineering CMM Refurbishment (In-Progress)

- Refurbishing a coordinate measuring machine (CMM) for the department of Mechanical Engineering, running Renishaw MODUS software, to restore it to working, measurement-ready condition.
- Diagnosing faults across the air supply (bearing), controller communication, probe, and scales, and carrying out repairs to restore motion and measurement.
- Planning to qualify the probe with the machine's reference sphere, then measure machined components to verify accuracy before releasing it for student projects.
- Documenting the diagnostic steps, repairs, and qualification procedure so future students and technicians can maintain and use the machine.

